Analyzing Change in the Real World

Dylan Fan
Jiawei Li
Yidou Zhou
Dylan Nguyen

1. Introduction

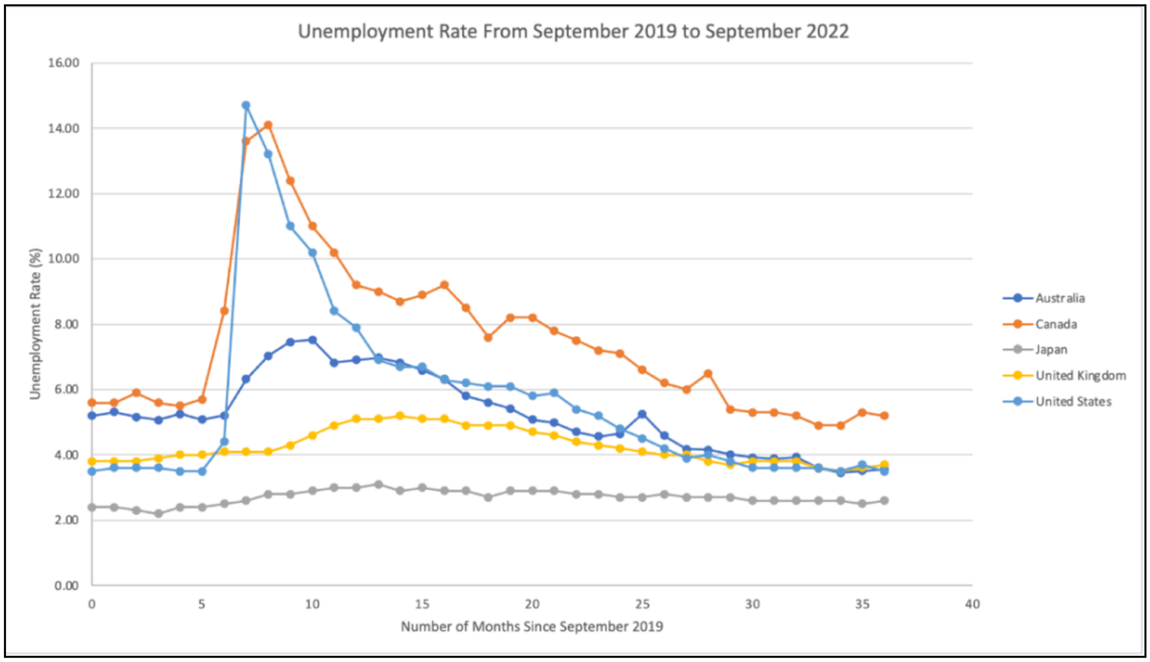
a) How does Covid-19 affect unemployment rates in various countries? The covid 19 pandemic had a profound global impact. It directed focus to the interconnectedness of the world and the state of public health services. Most importantly, it emphasized the lack of stability and resilience of the economy. To understand the consequential impact on the economy, several factors must be considered. This project will solely examine countries' unemployment rates over an extended period of time to determine the pandemic's impact.

Months since Sept. 2019	AUS Unemployment Rate (%)	Canada's Unemployment Rate (%)	Japan's Unemployment Rate(%)	UK's Unemployment Rate (%)	US's Unemployment Rate (%)
0	5.20	5.60	2.40	3.80	3.50
1	5.31	5.60	2.40	3.80	3.60
2	5.16	5.90	2.30	3.80	3.60
3	5.07	5.60	2.20	3.90	3.60
4	5.26	5.50	2.40	4.00	3.50
5	5.08	5.70	2.40	4.00	3.50
6	5.21	8.40	2.50	4.10	4.40
7	6.32	13.60	2.60	4.10	14.70
8	7.04	14.10	2.80	4.10	13.20
9	7.46	12.40	2.80	4.30	11.00
10	7.53	11.00	2.90	4.60	10.2

Figure 1.1

b) Figure 1.1 illustrates unemployment rates from September 2019 to July 2020 (10 rows of data). The table of data includes unemployment rates (2nd, 3rd, 4th, 5th, 6th colom) in percentages, time (1st colom) in months, and countries (Australia, Canada, Japan, United Kingdom, United States) differentiated by color. The independent/input variable is time and the dependent/output variable is the unemployment percentage. This data set was obtained on https://data.oecd.org/unemp/unemployment-rate.htm. The Organization for Economic Cooperation and Development collected unemployment rate percentage estimates to provide a data set that is internationally comparable. The organization achieved this through uniform application of the definition of unemployment and the access to labor force surveys.

2. Visual Representation



As appeared in the graph, there was a sharp increase in the unemployment rates for the US, Canada and Australia in April 2020. Unemployment in Canada and the United States soared and reached their peaks soon after (when jobs faced the biggest hit from the pandemic), while unemployment in Australia continued to rise during the following months.On the other hand, unemployment in the UK began to hike a few months later than the previous three, but recovered rather slowly. Out of the five countries, COVID had the least damage on Japan's job market, as the unemployment rate in Japan remained low and fairly steady throughout the 37 months.

3. Rates of Change

<u>Calculation of an average rate of change from September 2019 to October 2019 in Australia:</u>

$$Average\ rate\ of\ change\ = \frac{\triangle Unemployment\ Percentage}{\triangle months} = \frac{5.31-5.2}{1-0} = 0.11 = 11\%/\text{month}$$

The average rate of change of 0.11 means that from September 2019 to October 2019 in Australia, there was an increase in the unemployment rate by 11%/month.

<u>Calculation of instantaneous rate of change in September 2020 in Australia:</u>

$$Instantaneous\ rate\ of\ change\ = \frac{6.97-6.91}{13-12} = 0.06 = 6\%/\text{month}$$

The instantaneous rate of change of 0.06 means the change in the rate at a particular point in time, in this case, at September 2020. In other words, for every 1 month increase at Sept 2020, the unemployment in Australia would increase by 6 percent.

<u>Calculation of relative rate of change in September 2020 in Australia:</u>

$Relative\ rate\ of\ change = \frac{f'(September\ 2020)}{f(September\ 2020)} = \frac{0.06}{6.91} = 0.009 = 0.9\%$

The relative rate of change is the rate of change at its original value described as a percentage. In this case, in September 2020, would change 0.9% per month to October 2020.

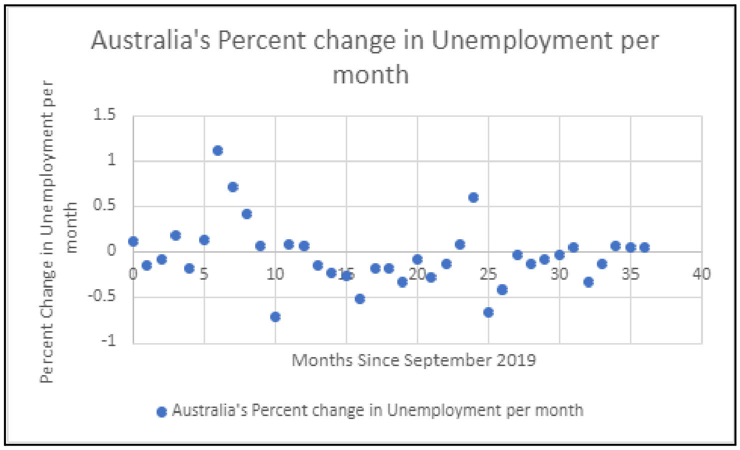
<u>Data table consisting of (approximate) values of the instantaneous rates of change:</u>

In order to calculate for the instantaneous rate of change, I used the right-hand method, taking the average rate of the closest point to the right of the original:

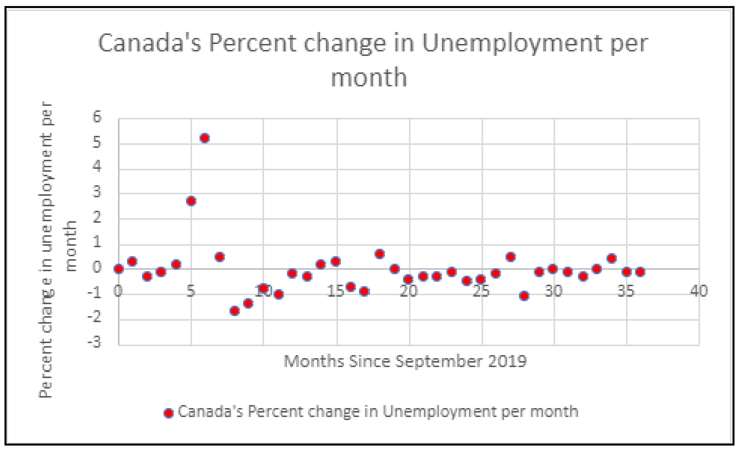
Example Calculation for September 2019: $\frac{5.31-5.2}{1-0} = 0.11$

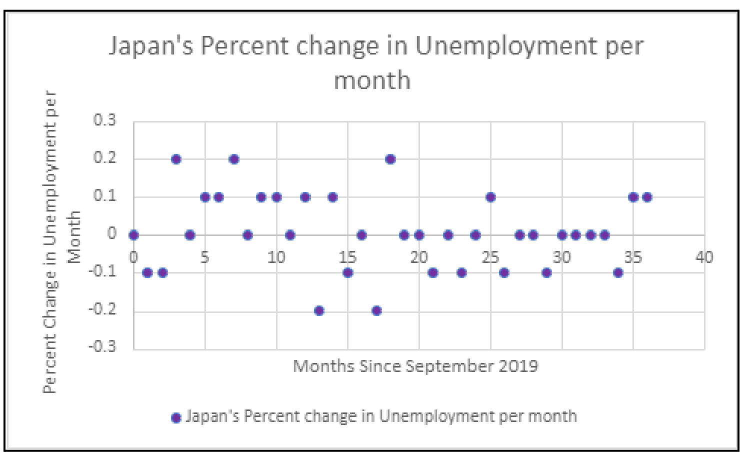
Months since Sept. 2019	AUS Unemployment Rate (%)	Instantaneous Rate of Change in AUS Unemployment Rate (%) per Month
0	5.2	0.11
1	5.31	-0.15
2	5.16	-0.09
3	5.07	0.19
4	5.26	-0.18
5	5.08	0.13
6	5.21	1.11
7	6.32	0.72
8	7.04	0.42
9	7.46	0.07
10	7.53	-0.71

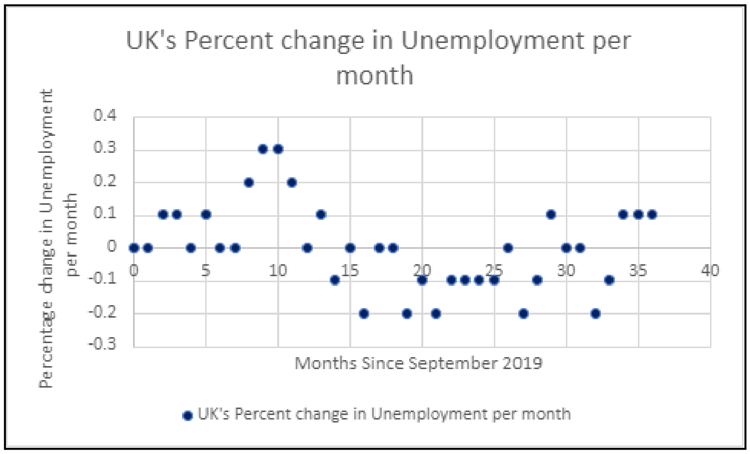
Using the Data found above along with the rest of the unemployment rate in Australia, we created a graph to represent the derivative of our function:

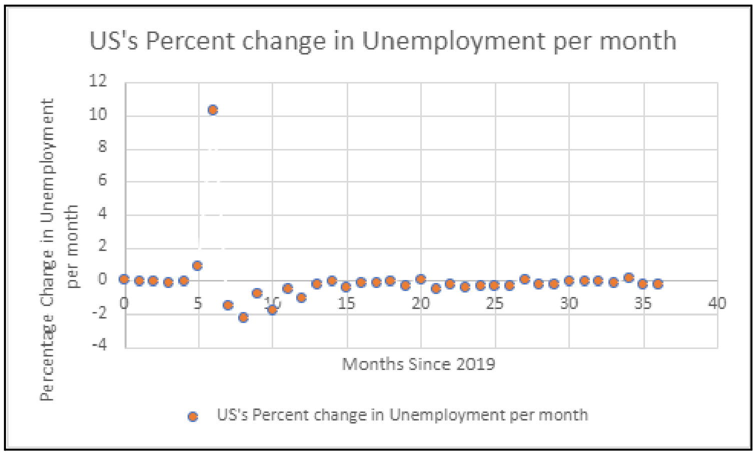


We also created a graph for the other four countries that represents the derivative of our function:









4. Conclusion:

In five nations between September 2019 and July 2020, our study examined the effects of COVID-19 on unemployment rates. Data showed considerable increases in unemployment during the pandemic, especially in the US, Canada, and Australia. These increases coincided with the global spread of COVID-19. The UK saw a delayed increase, while Japan continued to have a stable work market. For the majority of the countries, there is a large increase in the percent change in unemployment during the early 5-10 months since September 2020. This is equivalent to when looking at the non-derivative graph as at points from 5-10 months (since September 2020), those months have the greatest slopes.

This is relevant to the real world as during the early months of 2020, Covid-19 was becoming more prevalent and rapidly increasing, which led to an increase in layoffs and unemployment, as depicted from the graphs. This relates to our research question: How did Covid-19 affect unemployment in different countries, as we can clearly see from the graph, that there was a great increase in unemployment during the peak of Covid-19.

Reference

OECD. (n.d.). *Unemployment - unemployment rate - OECD data*. theOECD. https://data.oecd.org/unemp/unemployment-rate.htm